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**Assigned Organisation: North-West Cricket
Union(Potchefstroom)**

Industry: Sports



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Introduction

The following paper provides a description of the proposed network design for the North-West Cricket Union (Potchefstroom), which is to be done as a part of the CMPG325 Computer Networks semester project. The aim of the project is to build a network infrastructure that will meet the needs of communication defined in the assigned client scenario.

The following design is realized and simulated with the help of Cisco Packet Tracer software tool. It includes the design of the physical and logical topologies of the network, VLANs, and an IP addressing scheme based on the 192.168.51.0/24 address range. "Router-on-a-Stick" topology is used for inter-VLAN routing purposes.

The current Milestone 1 submission highlights the initial network planning and design process. This includes the client's specifications, physical and logical topologies, the suggested IP addressing approach, basic design considerations, and the first GitHub repository that will be used to track the progress of the work.

1. Client Requirements

1.1 Client Overview

This network design project is for the Northwest Cricket Union based in Potchefstroom. The company operates in the sports sector, and it requires a scalable, reliable and secure computer network to support departmental connectivity and communication. The network solution will be designed and simulated on Cisco Packet Tracer. The address block 192.168.51.0/24 has been assigned to the project.

1.2 Network Requirements

The North-West Cricket Union requires a network that meets the following requirements:

1. The network should support 5 separate departments or network groups.
2. The departments need to be logically divided for network administration, security and organization.
3. VLANs (Virtual Local Area Networks) would be used to separate the departments into different broadcast domains.
4. The network must use Router-on-a-Stick, also known as Inter-VLAN Routing, to allow controlled communication between the different VLANs.
5. The network must use the IP address block 192.168.51.0/24
6. Design a suitable IP addressing scheme to optimize the use of IP addresses for all necessary network devices.
7. The network needs to be built for CR5 to handle a 25% increase in users without renumbering the current network.

8. Printers should be shared across departments to allow authorized users of multiple departments to access required printing resources.
9. Departmental file sharing should not be allowed. Proper network security and access control mechanisms should be in place to prevent unauthorized access to files across departments.
10. The network shall ensure reliable connectivity between all authorized devices and network resources.
11. The network design must be scalable to allow for future growth of the organization.
12. Configure and test the finished network using Cisco Packet Tracer.
13. Necessary network constraints and successful connectivity must be tested and documented as proof.

1.3 Design Assumptions

The project brief does not state the number of users and or devices required for each department. Therefore, appropriate design assumptions will be documented during the network design phase.

The /27 network is assigned to the CR5 subnet with 30 valid host addresses. This gives reserve address capacity to handle the 25% user increase needed without renumbering.

The Cisco Packet Tracer simulation may use few representative end devices but the size of subnets chosen will be representative of the network's intended capacity.

2. Physical Topology

2.1. Overview

The physical topology is the real network devices and physical connections used in the Cisco packet tracer simulation. The design will include one router and one primary switch to keep the network simple and manageable. Each representative client device will connect to the main switch via Ethernet connections. Although the whole network is designed for more users, in the Cisco Packet Tracer simulation, less representative devices will be used.

2.2. Network Devices

The following devices will be used:

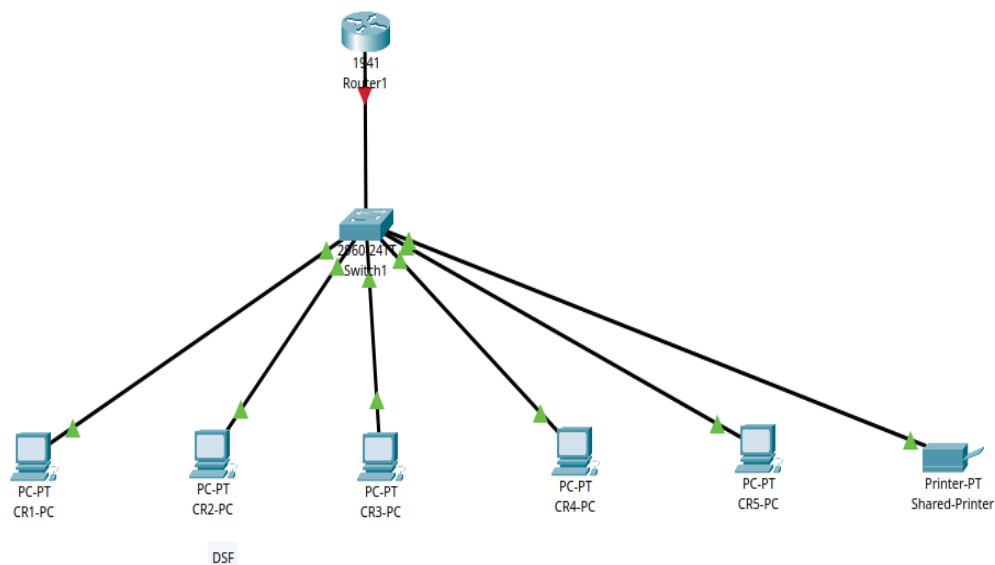
Device	Quantity	Purpose
Router	1	Provides Router-on-a-Stick and Inter-VLAN Routing
Switch	1	Connects all end devices and supports VLANs

PCs	5	One representative PC for each CR/VLAN
Printer	1	Shared printing resource

2.3. Physical Connections

The link between the router and the main switch would be done using Ethernet. All PCs and printer would be linked to the main switch using the Ethernet cables. In the later phase, a trunk link would be created between the router and the switch.

2.4. Physical Topology Diagram



2.5. Topology Selection and Justification

All end-devices of the proposed network are connected to the central switch (S1) using a star topology. Further, the switch connects to the router (R1). The router utilizes a Router-on-a-Stick technology, which will help in communicating between different VLANs.

The reason why a star topology has been selected for the proposed network is that it is simple and organized in nature and thus suitable for the Northwest Cricket Union. Since each end-device has its own link to the central switch, the network is easy to manage and troubleshoot. Other devices in the network can function properly even when an end-device link is down.

Extension of the topology is also very simple. The VLAN scheme allows different client groups to remain conceptually isolated from one another but enables adding additional devices to available switch ports. This means that the network can fulfill all the present demands of the client while giving room for future expansion, which includes CR5 expansion.

2.6. Switch Port Allocation

The following ports can be used to keep the configuration organised:

Switch Port	Connected Device	VLAN
G0/1	Router R1	Trunk
F0/1	CR1-PC1	10
F0/3	CR2-PC2	20
F0/5	CR3-PC1	30
F0/7	CR4-PC2	40
F0/9	CR5-PC1	50
F0/11	Shared Printer	60

2.7. Cable Types

The following cable connections will be used:

Router to Switch: Copper Straight-Through cable.

Switch to PCs: Copper Straight-Through cables.

Switch to Printer: Copper Straight-Through cable.

3. Logical Topology

3.1. Overview

A connection between the router and the primary switch will be made through Ethernet. The five network segments are logically split into separate Virtual Local Area Network (VLANs). Every VLAN is considered to be a unique broadcast domain and a logical network. In order to provide controlled connectivity between the VLANs, router-on-a-stick topology will be implemented. There will be one physical link between the router and the switch configured as a trunk link. Every VLAN will have its default gateway which is the router's subinterface. The shared printer will also be added to a different VLAN.

3.2. VLAN Allocation

Network Group	VLAN ID	VLAN Name	Subnet
CR1	10	CR1	192.168.51.0/27
CR2	20	CR2	192.168.51.32/27

CR3	30	CR3	192.168.51.64/27
CR4	40	CR4	192.168.51.96/27
CR5	50	CR5	192.168.51.128/27
Shared Printer	60	PRINTERS	192.168.51.160/27

3.3. Router-on-a-Stick Configuration Design

The router will connect to the main switch using one physical Ethernet interface.

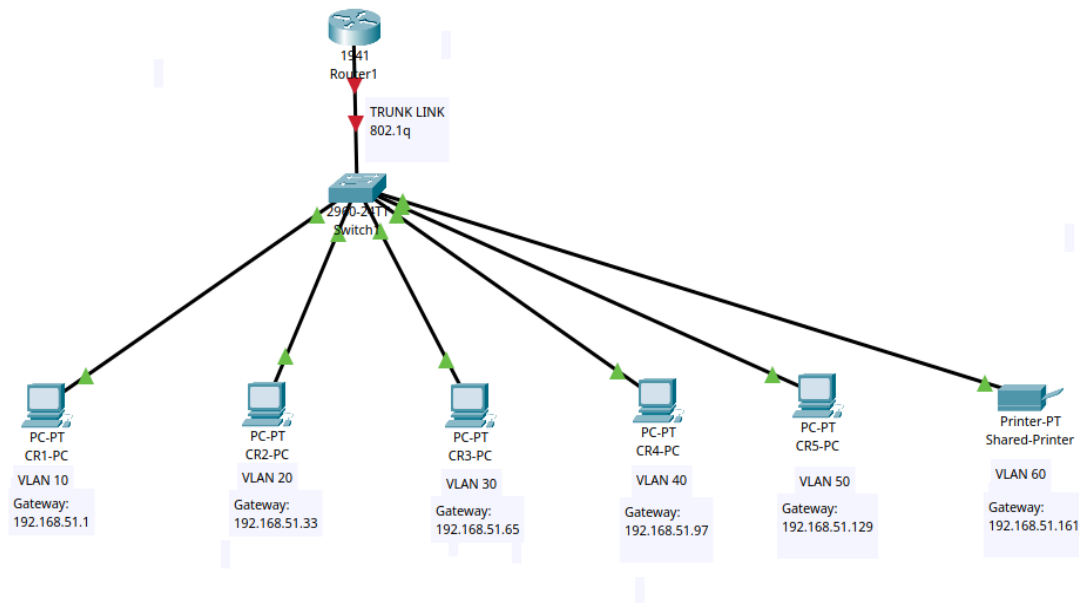
The physical router interface will be divided into the following logical subinterfaces:

G0/0.10 — VLAN 10 — CR1
G0/0.20 — VLAN 20 — CR2
G0/0.30 — VLAN 30 — CR3
G0/0.40 — VLAN 40 — CR4
G0/0.50 — VLAN 50 — CR5
G0/0.60 — VLAN 60 — Shared Printer

Each subinterface will have an IP address that acts as the default gateway for devices in that VLAN.

3.4. Logical Network Structure

The logical network will operate as follows:



3.5. Inter-VLAN Communication

VLAN segmentation is done at layer 2 level. Therefore, the router becomes essential for communication between different VLANs. In case of allowed communication, the "Router on a Stick" topology will provide the inter-VLAN routing.

Network will be configured in such a way that:

- Communication can occur between the devices that belong to the same VLAN.
- Authorized users from all departments will have access to the common printer.
- Necessary network configuration will prevent unauthorized file sharing among different departments.
- VLAN segmentation will be used for department separation.

4. IP Addressing Plan

4.1. Assigned Address Block

The North-West Cricket Union has been assigned the following address block:

192.168.51.0/24

This provides a total of 256 addresses, ranging from:

- Network address: 192.168.51.0
- Usable host addresses: 192.168.51.1 to 192.168.51.254
- Broadcast address: 192.168.51.255

To keep the network design simple and easy to manage, the address block will be divided into /27 subnets.

A /27 subnet provides:

- 32 total addresses
- 30 usable host addresses
- Subnet mask: 255.255.255.224

4.2. CR5 Growth Requirement

CR5 needs to be able to handle a 25% growth in number of users without re-numbering.

For the purposes of designing the network, CR5 will be capable of handling 20 users initially.

A 25% growth would mean:

$$20 \text{ users} \times 25\% = 5 \text{ users}$$

Thus, post growth, CR5 will have to accommodate:

$$20 + 5 = 25 \text{ users}$$

/27 subnet gives 30 usable addresses, which means that CR5 is capable of accommodating 25 users along with network devices without re-numbering.

4.3. VLAN and IP Address Allocation

Network Group	VLAN	Network Address	Subnet Mask	Gateway	Usable Host Range	Broadcast Address
CR1	10	192.168.51.0	255.255.255.224	192.168.51.1	192.168.51.1 – 192.168.51.30	192.168.51.31
CR2	20	192.168.51.32	255.255.255.224	192.168.51.33	192.168.51.33 – 192.168.51.62	192.168.51.63
CR3	30	192.168.51.64	255.255.255.224	192.168.51.65	192.168.51.65 – 192.168.51.94	192.168.51.95
CR4	40	192.168.51.96	255.255.255.224	192.168.51.97	192.168.51.97 – 192.168.51.126	192.168.51.127
CR5	50	192.168.51.128	255.255.255.224	192.168.51.129	192.168.51.129 – 192.168.51.158	192.168.51.159
Shared Printer	60	192.168.51.160	255.255.255.224	192.168.51.161	192.168.51.161 – 192.168.51.190	192.168.51.191

4.4. Device Addresses

The first usable address in each subnet will be used as the default gateway.

CR1 – VLAN 10

Gateway: 192.168.51.1

IP: 192.168.51.2

Mask: 255.255.255.224

CR2 – VLAN 20

Gateway: 192.168.51.34

IP: 192.168.51.34

Mask: 255.255.255.224

CR3 – VLAN 30

Gateway: 192.168.51.65

IP: 192.168.51.66

Mask: 255.255.255.224

CR4 – VLAN 40

Gateway: 192.168.51.97

IP: 192.168.51.98

Mask: 255.255.255.224

CR5 – VLAN 50

Gateway: 192.168.51.129

IP: 192.168.51.130

Mask: 255.255.255.224

Additional addresses from 192.168.51.132 to 192.168.51.158 remain available for current and future CR5 users.

Shared Printer – VLAN 60

Gateway: 192.168.51.161

IP: 192.168.51.162

4.5. Addressing Method

The network would be using static IP addressing in Cisco Packet Tracer. Depending upon the VLAN, each virtual PC would be assigned an IP address, subnet mask, and default gateway.

Router on Stick sub interfaces would be used by the router for each subnet's first usable address. Sub interfaces would act as default gateways for each VLAN.

Expansion of network in future would be easy due to availability of other addresses from 192.168.51.192 to 192.168.51.255.

5. Initial GitHub Repository

5.1. Repository Name

The following repository name will be used:

CMPG325-North-West-Cricket-Union-Network

The repository will be used to store the project documentation, Cisco Packet Tracer files, and evidence of the network design and testing.

Link:

<https://github.com/LeratoRamakatsa/CMPG325-North-West-Cricket-Union-Network.git>

5.2. Repository Structure

CMPG325-North-West-Cricket-Union-Network/

README.md

Documentation/

Client-Requirements.docx

IP-Addressing-Plan.docx

Logical-Topology.pkt

Physical-Topology.pkt

Packet-Tracer/

Evidence/

Screenshots/

Configs/

5.3. README File

The README file will provide a brief overview of the project.

It will include:

- Project name.
- Module code.
- Client name.
- Client location.
- Project purpose.
- Assigned IP address block.
- Network technologies used.
- Brief description of the repository structure.

5.4. Version Control

GitHub will be used to maintain a record of the project's development. Changes and additions to the project will be uploaded as the network is developed.

The repository will provide evidence of:

- Network planning.
- Network design decisions.
- Cisco Packet Tracer implementation.
- Testing and troubleshooting.
- Project progress.

6. References

- North-West University. (2026). *CMPG325 Computer Networks: Individual Semester Project Brief*. Potchefstroom: North-West University.
- Cisco Systems, Inc. (n.d.). *Cisco Packet Tracer Data Sheet*. Available at: [Cisco Packet Tracer Data Sheet](#) [Accessed 28 August 2026].
- Cisco Systems, Inc. (2025). *Configure Inter-VLAN Routing with the Use of an External Router*. Available at: [Cisco's Router-on-a-Stick guide](#) [Accessed 28 August 2026].
- Cisco Systems, Inc. (2026). *VLAN Configuration Guide: Configure VLAN Trunks*. Available at: [Cisco VLAN trunk configuration guide](#) [Accessed 28 August 2026].